

A Smart Employability Assessment and Curriculum Alignment System

Curriculum–Industry Skill Gap Analysis and Personalized Student Coaching

Department: Computer Science & Engineering | **Focus:** AI, Educational Technology & Automated Skill Assessment

1. BACKGROUND

The software and technology sector is undergoing an extraordinarily rapid evolution driven by breakthroughs in Artificial Intelligence, Machine Learning, Cloud Infrastructure, Cybersecurity, Data Science, DevOps practices, Full-Stack Web Technologies, and the Internet of Things (IoT). Modern IT employers increasingly expect Computer Science & Engineering (CSE) graduates to demonstrate immediate operational capability, practical tool fluency, software engineering best practices, and active exposure to modern development workflows right from their first day on the job.

However, higher education institutions operate under a fundamental structural constraint: university curriculum updates are constrained by academic governance, faculty senate reviews, and administrative compliance procedures that typically recur on a 3-to-5-year cycle. Conversely, industry software frameworks, cloud tools, continuous integration pipelines, and AI libraries update on a quarterly or monthly basis.

Because of this structural mismatch, CSE students spend four years acquiring foundational theoretical concepts, but when campus placement drives begin, they encounter a significant disconnect between what was taught in the classroom and what companies actively look for in technical interviews (such as container orchestration, modern API development, production AI deployment, and cloud security). While universities recognize this gap, traditional manual syllabus reviews across dozens of courses are extremely labor-intensive, time-consuming, and frequently reliant on anecdotal opinions rather than objective, real-time talent market analytics.

2. PROBLEM STATEMENT

Catchy Problem Statement:

"Computer Science academic programs struggle to keep pace with how fast the technology industry moves. Engineering graduates invest four years acquiring academic credentials, only to discover their practical skill sets are out of date for campus placements. We propose an AI-driven decision system that continuously audits university syllabi against real-time job market requirements, pinpoints exact skill deficiencies, and delivers customized, step-by-step learning roadmaps to ensure students become genuinely placement-ready."

Higher education institutions currently lack an automated, continuous mechanism to verify whether taught course outcomes align with active industry hiring requirements. Standard syllabus revisions take place manually every few years, which means critical emerging tools—such as Generative AI SDKs, Infrastructure as Code (IaC), automated testing frameworks, and cloud-native security—are often omitted entirely or taught only as high-level theoretical concepts without practical laboratory exposure.

Consequently, graduates face notable difficulty passing technical coding assessments and domain interviews during campus recruitment drives. At the same time, corporate employers are forced to allocate substantial time and capital

into post-hire bootcamp training programs to bridge these basic practical gaps. There is an urgent necessity for an intelligent, data-driven **Curriculum–Industry Skill Alignment System** that systematically ingests job postings, compares them against academic course syllabi, identifies missing technical competencies, calculates individual student placement readiness, and prescribes targeted learning pathways.

2.1 Existing System Problems

- **Curriculum–Industry Disconnect:** University syllabi updated infrequently create systemic blind spots in fast-evolving technical areas, leaving graduates without hands-on experience in modern industry tools.
- **Inadequate Employability Assessment:** Academic evaluation systems rely heavily on Grade Point Average (GPA) and written theoretical examinations. These metrics measure academic recall but fail to assess practical coding ability, software design skills, project execution, or tool mastery.
- **Lack of Granular, Personalized Skill Analysis:** Students receive broad letter grades without specific diagnostic feedback outlining which technical or professional skills they lack for specific job roles (e.g., Backend Developer vs. Data Analyst).
- **Static & Manual Curriculum Reviews:** Academic departments lack automated analytical tools to evaluate syllabus depth, topic coverage, and practical laboratory allocation against live job market data.
- **Generic Career Preparation:** Career guidance provided by placement cells is frequently one-size-fits-all rather than tailored to distinct engineering specialization tracks.
- **Reduced Graduate Placement & High Onboarding Costs:** The widening gap between classroom learning and employer expectations lowers recruitment conversion rates and forces corporations to spend heavily on retraining fresh hires.

2.2 Identified Research Gaps

A review of existing literature and recruitment tools highlights four primary limitations:

- Most existing systems focus purely on binary placement prediction (Placed / Not Placed) using historical GPA data, without explaining *why* a student is at risk or *how* to fix specific skill gaps.
- They lack automated Natural Language Processing (NLP) text-mining mechanisms to dynamically compare real job descriptions against course syllabus topics.
- They evaluate student academic records, coding profiles, projects, and certifications in separate silos rather than synthesizing them into a unified decision framework.
- They do not provide actionable, continuous feedback loops that serve both individual students (coaching) and university decision-makers (syllabus restructuring).

3. WHY THIS PROJECT MATTERS

The proposed framework delivers tangible value across all key higher education stakeholders:

- **For Students (Targeted Placement Prep):** Eliminates guesswork during job preparation by providing a clear, personal diagnostic report. Instead of studying generic material, students receive a step-by-step roadmap focusing on the exact missing skills, projects, and practice topics required for their dream role.
- **For Faculty & Syllabus Committees (Data-Driven Decisions):** Replaces subjective opinions with objective, live job market data when revising syllabi, introducing new electives, or adding modern software tools to lab exercises.

- **For Institutions & Accreditation (Quality Assurance):** Helps universities satisfy outcome-based education (OBE) criteria for accreditation bodies (such as NBA and NAAC) by proving continuous alignment between academic curriculum and industry demand.

Academic Suitability: As a B.Tech final-year capstone project, this work combines Natural Language Processing (NLP), Machine Learning classification, Data Analytics, and Web Application Development to build a real-world decision support system with immediate institutional utility.

4. KEY OBJECTIVES

1. **Automated Job Market Ingestion Pipeline:** Build a continuous ingestion module that collects, cleans, and extracts required technical and soft skill taxonomies from real-world job descriptions across major CSE domains.
2. **Multi-Factor Skill Gap Engine:** Develop a multi-dimensional analytics engine to evaluate university course syllabi against active industry demand based on topic coverage, conceptual depth, and practical laboratory hours.
3. **Formulate Placement Readiness Index (PRI):** Create a comprehensive scoring algorithm that aggregates a student's academic marks, coding assessment scores, project portfolio quality, certifications, and soft skills into a standardized readiness score.
4. **Generate Personalized Learning Roadmaps:** Build an intelligent recommendation engine that generates step-by-step study paths, recommended open-source capstone projects, and targeted certification suggestions tailored to individual student gaps.
5. **Detect & Flag Curriculum Anomalies:** Automatically flag critical syllabus defects, such as obsolete technologies being over-emphasized, high-demand industry skills missing entirely, or core courses restricted by low student intake limits.
6. **Deliver Executive Institutional Dashboards:** Design intuitive, interactive web dashboards for department heads and faculty to view domain health metrics, student cohort readiness, and data-backed syllabus revision proposals.

5. SCOPE OF THE PROJECT

What's Excluded (Out of Scope):

- Does not automatically alter official university transcript records or registrar databases without administrative authorization.
- Non-engineering academic disciplines (such as Arts, Commerce, or Basic Sciences) are excluded from the current design.
- Does not replace formal classroom instruction, faculty lectures, or official university term examinations.

What's Included (In Scope):

- Focuses on Undergraduate Computer Science & Engineering (CSE) and Information Technology (IT) programs.
- Covers 8 core technology domains: AI & Data Science, Cloud & DevOps, Full-Stack Web Development, Cybersecurity, Software Engineering, Mobile/Embedded Systems, CS Theory, and Soft Skills.
- System benchmark testing using a comprehensive curated dataset of 1,000 course and skill data points.
- Generation of individualized Student Diagnostic Reports, Skill Gap Matrices, and Institutional Curriculum Revision Proposals.

6. HOW IT WORKS (PROPOSED METHODOLOGY)

The proposed system operates across four structured operational phases using Natural Language Processing (NLP) text processing and Machine Learning scoring models:

Phase	Core Operational Process	Technique / Algorithm	Key Output / Deliverable
Step 1: Data Gathering	Extract course syllabus text (topics, credits, lab hours) and pull active industry job descriptions across CSE domains.	Regex Parsing, TF-IDF, Web Scraping, PDF Extraction.	Cleaned Academic Master Dataset & Live Skill Demand Feed.
Step 2: Skill Gap Calculation	Map syllabus concepts to job market requirements. Calculate coverage depth, alignment scores, and flag outdated topics.	Cosine Similarity, Sentence-BERT Embeddings, Rule Engine.	Multi-Factor Gap Analysis Engine Matrix & Red Flags.
Step 3: Student Diagnostic	Evaluate student quiz scores, coding profiles, projects, and grades. Compute Placement Readiness Index (PRI) and generate roadmap.	Weighted Scoring Logic, Random Forest Classifier.	Personalized Student Diagnostic & Custom Learning Roadmap.
Step 4: Department Summary	Aggregate cohort skill scores by domain (e.g., Web Dev, AI) to provide macro insights for department heads and syllabus committees.	Data Analytics Visualization, Priority Matrix Generation.	Executive Analytics Dashboard & Action Roadmap.

6.1 Mathematical Formulation of Placement Readiness Index (PRI)

To ensure an objective evaluation, the student Placement Readiness Index (*PRI*) for a target job role *R* is calculated as a weighted aggregation of five multi-dimensional skill vectors:

$$PRI_R = (w_1 \cdot S_{Academic}) + (w_2 \cdot S_{TechSkill}) + (w_3 \cdot S_{Projects}) + (w_4 \cdot S_{Certifications}) + (w_5 \cdot S_{SoftSkills})$$

Where $w_1 + w_2 + w_3 + w_4 + w_5 = 1.0$, representing role-specific weight parameters tuned for target job roles.

7. TIMELINE & EXECUTION PLAN

The project is scheduled across a structured 16-week timeline with defined bi-weekly milestones:

Duration	Main Focus & Phase	Key Activities & What Gets Delivered
Weeks 1–3	System Design & Requirements	Map system architecture, define database schemas, gather stakeholder requirements, and design UI wireframes. Deliverable: Architecture Document.
Weeks 4–6	Dataset Creation & Validation	Collect syllabi, scrape job descriptions, normalize text, and build the 1,000-sample benchmark dataset. Deliverable: Clean Benchmark Dataset.
Weeks 7–10	Gap Engine & Scoring Logic	Implement NLP similarity matching, anomaly detection rules, and the PRI scoring algorithm. Deliverable: Working Core Analytical Engine.
Weeks 11–13	Student Roadmap Module	Build the personalized recommendation engine for custom study paths and project suggestions. Deliverable: Recommendation Module.
Weeks 14–15	Dashboard Interface & Integration	Develop interactive web dashboards for students and faculty. Connect backend scripts with frontend UI. Deliverable: Interactive Web Application.
Week 16	Testing, Evaluation & Final Report	Perform end-to-end system testing, evaluate prediction accuracy, compile project report, and prepare live demo. Deliverable: Final Project Report & Demo.

8. RESOURCE REQUIREMENTS

- **Hardware Requirements:** Standard laptop or development workstation (Multi-core CPU, 16 GB RAM, 50 GB available SSD storage).
- **Software & Programming Stack:** Python 3.10+ (`pandas`, `numpy`, `scikit-learn` for data processing and machine learning; `openpyxl` for spreadsheet ingestion; `nlTK` / `spacy` / `sentence-transformers` for NLP text processing).
- **Database & Vector Storage:** PostgreSQL or SQLite for relational academic records; ChromaDB or FAISS for vector embeddings of job descriptions and syllabi.

- **Frontend Dashboard UI:** Lightweight web framework (Streamlit / Flask / React with Tailwind CSS) for demonstration dashboards.

9. POTENTIAL RISKS & HOW WE MANAGE THEM

Identified Risk Event	Risk Level	Mitigation & Management Strategy
Unstructured / Messy Syllabus PDFs	Medium	Implement standardized PDF pre-processing scripts using regular expressions and layout cleaning before passing text to NLP models.
Evolving Industry Jargon & Synonyms	High	Utilize contextual vector embeddings (e.g., Sentence-BERT) instead of rigid keyword matching, allowing the engine to recognize that "React.js" aligns with "Frontend Frameworks".
Student Disengagement / Task Fatigue	Medium	Break recommended learning roadmaps into modular, weekly bite-sized micro-goals rather than overwhelming long-term resource lists.
Data Sparsity in Student Records	Low	Incorporate baseline domain weightings and fallback diagnostic quizzes to accurately infer missing skill proficiency metrics.

10. EXPECTED BENEFITS

- **For Students:** Clear, actionable self-awareness of personal strengths and missing job competencies; elimination of guesswork during placement prep; structured pathways to clear technical interviews.
- **For Universities & Faculty:** Objective, empirical data to justify syllabus updates, add modern lab tools, and streamline compliance with accreditation standards (NBA/NAAC).
- **For Industry Employers:** Direct access to engineering graduates trained in contemporary tools and workflows, leading to significantly lower post-hire onboarding and training expenses.